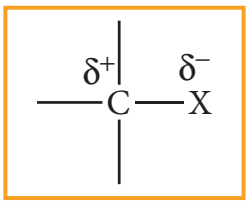
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The C –X bond is formed by overlap of sp^3 orbital of carbon atom with half filled p-orbital of the halogen atom. The atomic size of halogen increases from fluorine to iodine, which increases the C – X bond length. Larger the size, greater is the bond length, and weaker is the bond formed. The bond strength of C – X decreases from C – F to C – I in CH_3X. The changes in the value of bond length, bond enthalpy and bond polarity, as we more from C –F to C –I, is given in the table.

Table showing carbon – halogen bond length, bond enthalpy and polarity of bond.

Bond	Bond length (pm)	Bond Enthalpy ($kJmol^{-1}$)	Polarity of bond in terms of dipole moment (Debye)
$CH_3 - F$	139	452	1.847
$CH_3 - Cl$	178	351	1.860
$CH_3 - Br$	193	293	1.830
$CH_3 - I$	214	234	1.636

14.3.3 Methods of preparation

Haloalkanes are prepared by the following methods

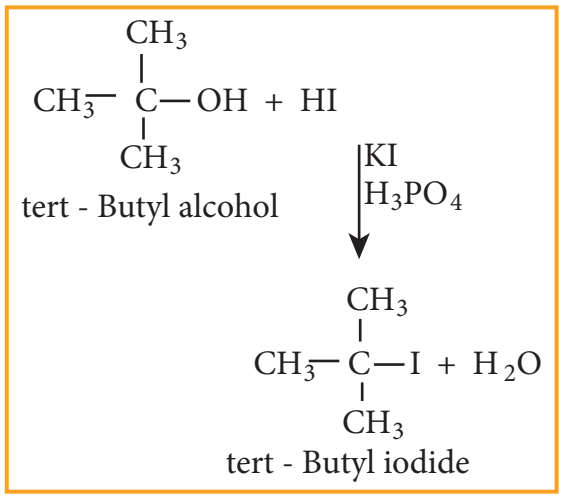
1) From alcohols

Alcohols can be converted into halo alkenes by reacting it with any one of the following reagent 1. hydrogen halide 2. Phosphorous halides 3. Thionyl chloride

a)Reaction with hydrogen halide

$$\underset{\text{Ethanol}}{CH_3CH_2OH} + HCl \xrightarrow[\Delta]{\text{Anhydrous } ZnCl_2} \underset{\text{Chloroethane}}{CH_3CH_2Cl} + H_2O$$

Mixture of con.HCl and anhydrous $ZnCl_2$ is called **Lucas reagent.**



The order of reactivity of halo acids with alcohol is in the order HI > HBr > HCl. The order of reactivity of alcohols with halo acid is tertiary > secondary > primary.

b)Reaction with phosphorous halides

Alcohols react with PX_5 or PX_3 to form haloalkane. PBr_3 and PI_3 are usually generated in situ (produced in the reaction mixture) by the reaction of red phosphorus with bromine and iodine, respectively.

Example

$$\underset{\text{Ethanol}}{CH_3CH_2OH} + PCl_5 \rightarrow \underset{\text{Chloro ethane}}{CH_3CH_2Cl} + POCl_3 + HCl$$

$$\underset{\text{Ethanol}}{3CH_3CH_2OH} + PCl_3 \rightarrow \underset{\text{Chloro ethane}}{3CH_3CH_2Cl} + H_3PO_3$$